| S.No: 1 | Exp. Name: ***Eldest of Three*** | **Date: 2026-07-25** |
|---|---|---|

**Aim:**

In a classroom, a teacher wants to identify the eldest student among the three. To simplify this task, the teacher decides to write a Python program that takes the ages of the three students and determines the oldest.

Can you help the teacher by writing this program?

**Input Format:**

- Three integers are entered on separate lines, each representing the age of a student.

**Output Format:**

- The output is the integer that represents the age of the elder student.

**Source Code:**

eldestAge.py

```
# Write your code here..
a = int(input())
b = int(input())
c = int(input())
#if a>=b and a>=c:
#       print(a)
#elif b>=a and b>=c:
#       print(b)
#else:
#       print(c)
print(max(a,b,c))
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 12 |
| 14 |
| 13 |


ID: 25B01A1216
2025-2029-IT-A
Shri Vishnu Engineering College for Women (Autonomous)

| 14 |
|---|

| Test Case - 2 |
|---|
| **User Output** |
| 11 |
| 43 |
| 23 |
| 43 |

| S.No: 2 | Exp. Name: ***Prime numbers from 2 to the given positive number*** | **Date: 2026-08-08** |
|---|---|---|

**Aim:**

Implement a program that takes an integer $n$ input from the user and prints all the prime numbers from 2 to $n$.

**Note:** A prime number is divisible only by 1 and itself.

**Input format:**

- The input contains a positive integer $n$.

**Output Format:**

- Contains the list of prime numbers up to $n$

**Example:**

**Input:**

10

**Output:**

[2, 3, 5, 7]

**Constraints:**

- The input n should be a positive integer greater than or equal to 2.
- The algorithm should be efficient, aiming for a time complexity of approximately $O(n * sqrt(n))$ where n is the input number.
- The algorithm should handle large inputs reasonably well, without consuming excessive memory or taking an unreasonable amount of time to execute.

**Source Code:**

CTP32252.py

```
n = int(input())

is_prime = [True] * (n + 1)
is_prime[0] = is_prime[1] = False

p = 2
while p * p <= n:
    if is_prime[p]:
        for i in range(p * p, n + 1, p):
            is_prime[i] = False
    p += 1

primes = [i for i in range(2, n + 1) if is_prime[i]]
print(primes)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| 10 |
| [2, 3, 5, 7] |

| Test Case - 2 |
| --- |
| **User Output** |
| 20 |
| [2, 3, 5, 7, 11, 13, 17, 19] |

| Test Case - 3 |
| --- |
| **User Output** |
| 30 |
| [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] |

| S.No: 3 | Exp. Name: ***Swap two numbers*** | Date: 2026-07-25 |
|---|---|---|

**Aim:**

Write a Python program that reads two numbers from the user and swaps their values using only two variables.

**Source Code:**

swapping.py

```
#Type Content here..
a=float(input("Enter the first number: "))
b=float(input("Enter the second number: "))
print("Original numbers:")
print("First number:",a)
print("Second number:",b)
#a,b=b,a
print("Swapped numbers:")
print("First number:",b)
print("Second number:",a)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| Enter the first number: |
| 2 |
| Enter the second number: |
| 3 |
| Original numbers: |
| First number: 2.0 |
| Second number: 3.0 |
| Swapped numbers: |
| First number: 3.0 |
| Second number: 2.0 |

| Test Case - 2 |
|---|
| **User Output** |

```
Enter the first number:
-3.5
Enter the second number:
2.5
Original numbers:
First number: -3.5
Second number: 2.5
Swapped numbers:
First number: 2.5
Second number: -3.5
```

| S.No: 4 | Exp. Name: ***Arithmetic Operations (Addition, Subtraction, Multiplication, Division)*** | **Date: 2026-07-25** |
|---|---|---|

**Aim:**

Write a program to read two positive integer inputs from the user and perform the following arithmetic operations: addition, subtraction, multiplication, and division, and print the result.

**Input Format**:

- The first two lines represent the input of two integers $num1$ and $num2$.

**Output Format:**

- The first line should print the sum of the provided inputs.
- The second line should print the subtraction of the provided inputs.
- The third line should print the multiplication of the provided inputs.
- The fourth line should print the division of the provided inputs.

**Sample Test Case:**

**Input:**

```
num1: 40
num2: 20
```

**Output:**

```
Addition of 40 and 20 = 60
Subtraction of 40 and 20 = 20
Multiplication of 40 and 20 = 800
Division of 40 and 20 = 2.0
```

**Note:**

- Assume the value of $num2$ is always greater than 0 (zero).

**Source Code:**

Arithexample2.py

```
# write the code...
a = int(input("num1: "))
b = int(input("num2: "))
print("Addition of",a,"and",b,"=",a+b)
print("Subtraction of",a,"and",b,"=",a-b)
print("Multiplication of",a,"and",b,"=",a*b)
print("Division of",a,"and",b,"=",a/b)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| num1: |
| 40 |
| num2: |
| 20 |
| Addition of 40 and 20 = 60 |
| Subtraction of 40 and 20 = 20 |
| Multiplication of 40 and 20 = 800 |
| Division of 40 and 20 = 2.0 |

| Test Case - 2 |
| --- |
| **User Output** |
| num1: |
| 4 |
| num2: |
| 6 |
| Addition of 4 and 6 = 10 |
| Subtraction of 4 and 6 = -2 |
| Multiplication of 4 and 6 = 24 |
| Division of 4 and 6 = 0.6666666666666666 |

| S.No: 5 | Exp. Name: ***Logical Operators*** | Date: 2026-07-25 |
|---|---|---|

**Aim:**

Write a Python program that reads two integers and performs logical operations using the `and`, `or`, and `not` operators.

**Input Format:**

- The first line contains an integer $num1$.
- The second line contains an integer $num2$.

**Output Format:**

- Print the results of the logical operations in the following format:

```
<num1> and <num2>: <result>
<num1> or <num2>: <result>
not <num1>: <result>
not <num2>: <result>
```

**Note:** Refer to the visible test cases for better understanding.

**Source Code:**

logicoperator.py

```
# Write your code here...
a = int(input())
b = int(input())
print(f"{a} and {b}: {a and b}")
print(f"{a} or {b}: {a or b}")
print(f"not {a}: {not a}")
print(f"not {b}: {not b}")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 3 |
| 4 |
| 3 and 4: 4 |

| |
|---|
| 3 or 4: 3 |
| not 3: False |
| not 4: False |

| Test Case - 2 |
|---|
| **User Output** |
| 0 |
| 1 |
| 0 and 1: 0 |
| 0 or 1: 1 |
| not 0: True |
| not 1: False |

| Test Case - 3 |
|---|
| **User Output** |
| 15 |
| 45 |
| 15 and 45: 45 |
| 15 or 45: 15 |
| not 15: False |
| not 45: False |

| S.No: 6 | Exp. Name: ***Comparison Operators*** | Date: 2026-07-25 |
|---|---|---|


ID: 25B01A1216

Shri Vishnu Engineering College for Women (Autonomous)

**Aim:**

Write a Python program that reads two integers and demonstrates the use of comparison operators by performing the six comparison operations (==, !=, <, <=, >, and >=).

**Input Format:**

- The first line contains an integer $num1$.
- The second line contains an integer $num2$.

**Output Format:**

- Print the results of all six comparison operations, each on a separate line in the following format:

```
<num1> == <num2> : <boolean result>
<num1> != <num2> : <boolean result>
<num1> < <num2> : <boolean result>
<num1> <= <num2> : <boolean result>
<num1> > <num2> : <boolean result>
<num1> >= <num2> : <boolean result>
```

**Source Code:**

operators.py

```
#write your code here
num1 = int(input())
num2 = int(input())

print(f"{num1} == {num2} : {num1==num2}")
print(f"{num1} != {num2} : {num1!=num2}")
print(f"{num1} < {num2}  : {num1<num2}")
print(f"{num1} <= {num2} : {num1<=num2}")
print(f"{num1} > {num2}  : {num1>num2}")
print(f"{num1} >= {num2} : {num1>=num2}")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 4 |
| 4 |

| |
|---|
| 4 == 4 : True |
| 4 != 4 : False |
| 4 < 4 : False |
| 4 <= 4 : True |
| 4 > 4 : False |
| 4 >= 4 : True |

| Test Case - 2 |
|---|
| **User Output** |
| 3 |
| 2 |
| 3 == 2 : False |
| 3 != 2 : True |
| 3 < 2 : False |
| 3 <= 2 : False |
| 3 > 2 : True |
| 3 >= 2 : True |

| Test Case - 3 |
|---|
| **User Output** |
| 65 |
| 76 |
| 65 == 76 : False |
| 65 != 76 : True |
| 65 < 76 : True |
| 65 <= 76 : True |
| 65 > 76 : False |
| 65 >= 76 : False |

| S.No: 7 | Exp. Name: ***Writing an example using identity operator "is not"*** | **Date: 2026-08-01** |
|---|---|---|

**Aim:**

Take two integers `x` and `y` as input from the console using `input()` function. Using the identity operator `is not` check if `x` and `y` are same objects or not, print to the console, the result of the two input integers as shown in the example.

**Sample Input and Output 1:**

```
x: -8
y: -8
-8 is not -8 True
```

**Sample Input and Output 2:**

```
x: 256
y: 256
256 is not 256 False
```

**Source Code:**

IdenopExample3.py

```
# Take user input for x
x = int(input("x: "))
# Take user input for y
y = int(input("y: "))
# check whether x is not y and print the result
print(x, 'is not', y,x is not y)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| x: |
| -8 |
| y: |
| -8 |


ID: 25B01A1216

Shri Vishnu Engineering College for Women (Autonomous)

| -8 is not -8 True |
|---|

| Test Case - 2 |
|---|
| **User Output** |
| x: |
| 256 |
| y: |
| 256 |
| 256 is not 256 False |

| S.No: 8 | Exp. Name: ***Bitwise Operator*** | Date: 2026-08-01 |
|---|---|---|

**Aim:**

Take two integers x and y as inputs from the console using input() function. For each bitwise operator ( >> , <<, &, |, ~, and ^ ), print to the console, the result of applying these operators on the two input integers as shown in the example.

**Sample Input and Output:**

```
x: 52
y: 20
52 >> 20 is 0
52 << 20 is 54525952
52 & 20 is 20
52 | 20 is 52
~ 52 is -53
52 ^ 20 is 32
```

**Source Code:**

Bitopexample1.py

```
#Type Code here...
x = int(input("x: "))
y = int(input("y: "))
print(f"{x} >> {y} is {x>>y}")
print(f"{x} << {y} is {x<<y}")
print(f"{x} & {y} is {x&y}")
print(f"{x} | {y} is {x|y}")
print(f"~ {x} is {~x}")
print(f"{x} ^ {y} is {x^y}")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| x: |
| 30 |
| y: |
| 2 |
| 30 >> 2 is 7 |
| 30 << 2 is 120 |

| |
|---|
| 30 & 2 is 2 |
| 30 \| 2 is 30 |
| ~ 30 is -31 |
| 30 ^ 2 is 28 |

| Test Case - 2 |
|---|
| **User Output** |
| x: |
| 10 |
| y: |
| 20 |
| 10 >> 20 is 0 |
| 10 << 20 is 10485760 |
| 10 & 20 is 0 |
| 10 \| 20 is 30 |
| ~ 10 is -11 |
| 10 ^ 20 is 30 |

| S.No: 9 | Exp. Name: ***Write a program on assignment operators =,+=,-= and *=*** | **Date: 2026-08-01** |
|---|---|---|

**Aim:**

Take two integers `x` and `y` as inputs from the console using `input()` function. For each assignment operator ( `=` , `+=` , `-=` , and `*=` ), print to the console, the result of applying these operators on the two input integers as shown in the example.

Assumption:

- y is a non-zero integer

**Sample Input and Output:**

```
x: 20
y: 30
x = y: 30
x += y: 50
x -= y: -10
x *= y: 600
```

**Hint**: Before every assignment operation, reset the values of x and y to the initial values.

**Source Code:**


ID: 25B01A1216

Shri Vishnu Engineering College for Women (Autonomous)

Assignexample2.py

```
# Assignment Operators =, +=, -=, *=
x = int(input("x: "))
y = int(input("y: "))
a=x
b=y
#x=y
x=a
y=b
x=y
print("x = y:", x)
#x+=y
x=a
y=b
x+=y
print("x += y:", x)
#x-=y
x=a
y=b
x -= y
print("x -= y:", x)
#x*=y
x=a
y=b
x *= y
print("x *= y:", x)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| x: |
| 20 |
| y: |
| 30 |
| x = y: 30 |
| x += y: 50 |
| x -= y: -10 |
| x *= y: 600 |

| Test Case - 2 |
| --- |
| **User Output** |
| x: |
| 6 |
| y: |
| 3 |
| x = y: 3 |
| x += y: 9 |
| x -= y: 3 |
| x *= y: 18 |

| S.No: 10 | Exp. Name: ***Ternary Operator*** | Date: 2026-08-01 |
|---|---|---|


ID: 25B01A1216

Shri Vishnu Engineering College for Women (Autonomous)

**Aim:**

Write a python program to find the maximum value of given two numbers using ternary operator.

**Input Format:**

- First line should prompt the user to enter first integer
- Second line should prompt the user to enter second integer

**Output Format:**

- Output should print the maximum value using ternary operator

**Source Code:**

ternary.py

```
x = int(input())
y = int(input())
max_value = x if x>y else y
''' write your code here using ternary operator'''
print(max_value)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 91 |
| 123 |
| 123 |

| Test Case - 2 |
|---|
| **User Output** |
| 15 |
| -23 |
| 15 |

| S.No: 11 | Exp. Name: ***Membership Operator*** | Date: 2026-08-01 |
|---|---|---|


ID: 25B01A1216

Shri Vishnu Engineering College for Women (Autonomous)

**Aim:**

Write a Python program to demonstrate the use of the membership operators `in` and `not in` with a predefined list of integers.

**Input Format:**

- A single line contains an integer $n$ to be checked for membership in the predefined list.

**Output Format:**

- Print "True" or "False" indicating whether $n$ is present in the list using the in operator.
- Print "True" or "False" indicating whether $n$ is not present in the list using the not in operator.

**Note:**

- The predefined list of integers is already provided in the code.
- Refer to the visible test cases for better understanding.

**Source Code:**

membership.py

```
num_list = [1, 2, 3, 4, 5, 15, 25, 89, 54, 123, 489]
# Write your code here...
n = int(input())
print(n in num_list)
print(n not in num_list)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 21<br>False<br>True |

| Test Case - 2 |
| --- |
| **User Output** |
| -3 |
| False |
| True |

| Test Case - 3 |
| --- |
| **User Output** |
| 5 |
| True |
| False |

| Test Case - 4 |
| --- |
| **User Output** |
| 54 |
| True |
| False |

| S.No: 12 | Exp. Name: ***Arithmetic Operations on Complex Numbers*** | **Date: 2026-08-01** |
|---|---|---|

**Aim:**

Python provides built-in support for complex numbers. Once a complex number is created, it behaves like any other numeric data type, allowing you to perform arithmetic operations using standard operators.

A complex number can be entered using the `complex()` function.

**Syntax:**

```
complex(input())
```

When the user enters a value such as 3+4j or 10-2j, the complex() function converts the input string into a complex number.

**Example:**

```
c = complex(input("Enter a complex number: "))
print(c)
print(type(c))
```

**Sample Input:**

```
5+2j
```

**Output:**

```
(5+2j)
<class 'complex'>
```

Python supports all common arithmetic operators on complex numbers.

**Example:**

```
c1 = 2 + 3j
c2 = 1 + 4j
print(c1 + c2)
print(c1 - c2)
print(c1 * c2)
print(c1 / c2)
```

Python automatically computes the correct complex result for each operation.

# Task

Take two complex numbers as input from the console. For each arithmetic operator( +, -, *, and / ), perform these operations on the two complex numbers and print the result as shown in the example.

**Sample Input and Output:**

```
a1: 10+3j
b2: 14+3j
a1 + b2 = (24+6j)
a1 - b2 = (-4+0j)
a1 * b2 = (131+72j)
a1 / b2 = (0.7268292682926829+0.05853658536585366j)
```

**Source Code:**

ComplexNums.py

```
# Take input for the first complex number a1
c1 = complex(input("a1: "))
# Take input for the second complex number b2
c2 = complex(input("b2: "))

# perform + operation on the complex numbers and print the result
print("a1 + b2 =",c1 +c2)

# subtract b2 from a1 and print the result
print("a1 - b2 =",c1-c2)

# multiple a1, b2 and print the result
print("a1 * b2 =",c1*c2)

# divide b2 from a1 and print the result
print("a1 / b2 =",c1/c2)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| a1: |
| 10+3j |
| b2: |
| 41+3j |
| a1 + b2 = (51+6j) |
| a1 - b2 = (-31+0j) |
| a1 * b2 = (401+153j) |
| a1 / b2 = (0.24792899408284022+0.055029585798816574j) |

| Test Case - 2 |
|---|

| User Output |
| --- |
| a1: |
| 45+9j |
| b2: |
| 15+8j |
| a1 + b2 = (60+17j) |
| a1 - b2 = (30+1j) |
| a1 * b2 = (603+495j) |
| a1 / b2 = (2.58477508650519-0.7785467128027682j) |

| S.No: 13 | Exp. Name: ***Print Multiplication Table*** | Date: 2026-08-01 |
|---|---|---|






**Aim:**

Write a Python program to print the multiplication table of a given number in the given format.

**Input Format:**

- Read an integer $n$ with a prompt "Enter a number : ".

**Output Format:**

- Print the multiplication table of $n$ from 1 to 10.
- Each line should contain the value of $n$, the multiplier $i$, and the product in the following format:

`<n>*<i>=<result>`

where,$i$ ranges from 1 to 10, and $result$ represents $n \times i$.

**Note:** Refer to visible test cases for better understanding of input and output formats.

**Source Code:**

multiplication_table.py

```
# Type Content here...
n = int(input("Enter a number : "))
for i in range(1,11):
        print(f"{n}*{i}={n*i}")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| Enter a number : |
| 2 |
| 2*1=2 |
| 2*2=4 |
| 2*3=6 |
| 2*4=8 |
| 2*5=10 |
| 2*6=12 |

| |
|---|
| 2*7=14 |
| 2*8=16 |
| 2*9=18 |
| 2*10=20 |

| Test Case - 2 |
|---|
| **User Output** |
| Enter a number : |
| 13 |
| 13*1=13 |
| 13*2=26 |
| 13*3=39 |
| 13*4=52 |
| 13*5=65 |
| 13*6=78 |
| 13*7=91 |
| 13*8=104 |
| 13*9=117 |
| 13*10=130 |

| Test Case - 3 |
|---|
| **User Output** |
| Enter a number : |
| -10 |
| -10*1=-10 |
| -10*2=-20 |
| -10*3=-30 |
| -10*4=-40 |
| -10*5=-50 |
| -10*6=-60 |
| -10*7=-70 |
| -10*8=-80 |
| -10*9=-90 |
| -10*10=-100 |

| S.No: 14 | Exp. Name: ***Function with Multiple Return Values*** | **Date: 2026-08-29** |
|---|---|---|

**Aim:**

Write a Python program that defines a function **calculate_stats** to calculate multiple values from a given list of numbers. The function should return three values: the sum, the mean, and the maximum of the numbers.

**Input Format:**

- A single line containing space-separated integers representing the elements of the list

**Output Format:**

- The first line prints the sum of all the integers.
- The second line prints the mean (average) of the integers, formatted to two decimal places.
- The third line prints the maximum value among the integers.

**Constraints:**

- $0 < length\ of\ the\ list <= 100$

**Note:**

- You only need to write the code inside the **calculate_stats** function. The rest of the code is already provided

**Source Code:**

multiplefun.py

```
def calculate_stats(numbers):
        total=sum(numbers)
        mean=total/len(numbers)
        maximum=max(numbers)
        return total,mean,maximum
numbers = list(map(int, input().split()))
total, mean, maximum = calculate_stats(numbers)
print(total)
print(f"{mean:.2f}")
print(maximum)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| -23 -45 -56 -90 -12 -87 |
| -313 |
| -52.17 |
| -12 |

| Test Case - 2 |
| --- |
| **User Output** |
| 12 23 34 45 56 67 78 89 |
| 404 |
| 50.50 |
| 89 |

| S.No: 15 | Exp. Name: ***Default Arguments Function*** | Date: 2026-08-29 |
|---|---|---|

**Aim:**

Write a program to define a function **greet()** with default arguments. This function should accept three parameters: **name, age,** and **country**. If the user enters -1 for any input, the corresponding default value must be used. The default values are:

- **name** = "Dude"
- **age** = 0
- **country** = "Space"

You are also provided with a helper function **get_user_input()** that reads input from the user and returns the appropriate values after applying the default rules.

**Input Format:**

- First line prompts the user to enter the $name$ by displaying: "-1 for default: "
  - If -1 is entered, the default value "Dude" is used.
- Second line prompts the user to enter the $age$ by displaying: "-1 for default: "
  - If -1 is entered, the default value "0" is used.
- Third line of input should prompt the user to enter the $country$ by displaying: "-1 for default: "
  - If -1 is entered, the default value "Space" is used.

**Output Format:**

- Print the following line with the provided (or default) inputs:

```
Hello, {name}! You are {age} years old and you are from {country}
```

**Source Code:**


ID: 25B01A1216

Shri Vishnu Engineering College for Women (Autonomous)

default.py

```
# Define a function with default arguments
def greet(name="Dude",age=0,country="Space"):
        print("Hello, "+ name + "! You are "+ str(age) +" years old and
you are from "+ country)


# Get user input function
def get_user_input():
        name = input("-1 for default: ")
        if  name=="-1":
                name="Dude"
        age = input("-1 for default: ")
        if  age=="-1":
                age=0
        country = input("-1 for default: ")
        if  country=="-1":
                country="Space"
        return name,age,country




name, age, country = get_user_input()
greet(name, age, country)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| -1 for default: |
| -1 |
| -1 for default: |
| -1 |
| -1 for default: |
| -1 |
| Hello, Dude! You are 0 years old and you are from Space |

| Test Case - 2 |
| --- |
| **User Output** |
| -1 for default: |
| Ram |
| -1 for default: |
| -1 |
| -1 for default: |
| -1 |
| Hello, Ram! You are 0 years old and you are from Space |

| Test Case - 3 |
| --- |
| **User Output** |
| -1 for default: |
| Vikas |
| -1 for default: |
| 26 |
| -1 for default: |
| America |
| Hello, Vikas! You are 26 years old and you are from America |

| S.No: 16 | Exp. Name: ***Length of the String*** | Date: 2026-09-05 |
|---|---|---|

**Aim:**

Write a Python program to find the length of a string without using any library functions

**Input Format:**

- A single line containing a string $s$.

**Output Format:**

- A single integer representing the length of the input string.

**Source Code:**

length.py

```
def string_length(s):
        count=0
        for char in s :
                count+=1
        return count



input_string = input() # Taking user input
length = string_length(input_string) # Calculating length using the
function
print(length) # Printing the length
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| Code Tantra |
| 11 |

| Test Case - 2 |
|---|
| **User Output** |

| Hello@World!! |
| --- |
| 13 |

| Test Case - 3 |
| --- |
| **User Output** |
| 12345678 |
| 8 |

| S.No: 17 | Exp. Name: ***Count of Substring in String*** | Date: 2026-08-29 |
|---|---|---|

**Aim:**

Write a program to count the number of occurrences of a given substring within a string,.

**Input Format:**

- The first line contains a string representing the main string.
- The second line contains a string representing the substring to search for.

**Output Format:**

- Print a single integer representing the total number of occurrences of the substring in the main string.

**Constraints:**

- $1 \leq len(string) \leq 200.$

**Note:**

- The string comparison is case-sensitive.

**Source Code:**

SubstringCount.py

```
s1=input()
s2=input()
count=0
for i in range(len(s1)-len(s2)+1):
        if s1[i:i + len(s2)] == s2:
                count+= 1
print(count)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| ABCDCDC |
| CDC |
| 2 |

| Test Case - 2 |
| --- |
| **User Output** |
| CodeTantra |
| co |
| 0 |

| S.No: 18 | Exp. Name: ***List Operations*** | Date: 2026-09-05 |
|---|---|---|


ID: 25B01A1216

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**Aim:**

Write a program to perform the following operations on a list.

1. Addition (add an element to the list)
2. Insertion (insert an element at a specified index)
3. Slicing (extract a sublist using start and end indexes)

**Input Format:**

- The first line should take space-separated integers to form the initial list.
- The second line of input should take two space-separated integers, representing the index and value for the insertion.
- The third line should take two space-separated integers, representing the start index and end index for slicing.

**Output Format:**

- First line should display the initial list.
- Second line should display the list after insertion.
- Third line should display the list after slicing.

**Sample Input:**

```
1 2 3 4 5
0 5
0 3
```

**Sample Output:**

```
[1, 2, 3, 4, 5]
[5, 1, 2, 3, 4, 5]
[5, 1, 2]
```

**Note:**

- In Python, `append()` adds an element to the end of a list.
- To insert an element at a specific position, use the `insert()` method.

**Source Code:**

listOp.py

```
# Type Content here...
# Read the initial list
lst = list(map(int, input().split()))

# Read index and value for insertion
index, value = map(int, input().split())

# Read start and end indexes for slicing
start, end = map(int, input().split())

# Display initial list
print(lst)

# Insert the element at the specified index
lst.insert(index, value)

# Display list after insertion
print(lst)

# Display the sliced list
print(lst[start:end])
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 1 2 3 4 5 |
| 0 5 |
| 0 3 |
| [1, 2, 3, 4, 5] |
| [5, 1, 2, 3, 4, 5] |
| [5, 1, 2] |

| Test Case - 2 |
|---|
| **User Output** |
| 12 52 65 41 23 89 654 |
| 7 485 |

| 0 8 |
| --- |
| [12, 52, 65, 41, 23, 89, 654] |
| [12, 52, 65, 41, 23, 89, 654, 485] |
| [12, 52, 65, 41, 23, 89, 654, 485] |

| S.No: 19 | Exp. Name: ***Demonstration of Working with Lists*** | Date: 2026-09-05 |
|---|---|---|

**Aim:**

Write a program to demonstrate working with lists in python.

**Sample Test Case:**

**Input:**

Enter a list of elements (separated by commas): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

**Output:**

List of elements: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Length of list: 10
Minimum value: 1
Maximum value: 10
Sum of values: 55
Sorted list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

**Source Code:**

list.py

```
elements = input("Enter a list of elements (separated by commas): ")

lst = [int(x.strip()) for x in elements.split(",")]

print("List of elements:", lst)
print("Length of list:", len(lst))
print("Minimum value:", min(lst))
print("Maximum value:", max(lst))
print("Sum of values:", sum(lst))
print("Sorted list:", sorted(lst))
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| Enter a list of elements (separated by commas): |
| 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| List of elements: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] |

| Length of list: 10 |
| --- |
| Minimum value: 1 |
| Maximum value: 10 |
| Sum of values: 55 |
| Sorted list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] |

| Test Case - 2 |
| --- |
| **User Output** |
| Enter a list of elements (separated by commas): |
| 3, 8, 2, 9, 5 |
| List of elements: [3, 8, 2, 9, 5] |
| Length of list: 5 |
| Minimum value: 2 |
| Maximum value: 9 |
| Sum of values: 27 |
| Sorted list: [2, 3, 5, 8, 9] |

| S.No: 20 | Exp. Name: ***Tuple Creation*** | Date: 2026-09-21 |
|---|---|---|

**Aim:**

Write a Python program that prompts the user to enter details for two members and creates tuples for each member consisting of: name (string), age (int), address (string), college (string). Then, concatenate these two tuples into a single tuple and print all three tuples.

**Input Format:**

1. The first line contains four space-separated values for Member 1:
   - name
   - age
   - address
   - college
2. The second line contains the details of Member 2 in the same format

**Output Format:**

- Print the tuple for Member 1.
- Print the tuple for Member 2.
- Print the concatenated tuple of both members.

**Sample Input:**

```
John 12 Tampa St.Joseph
Jane 15 Florida St.Mary
```

**Sample Output:**

```
('John', 12, 'Tampa', 'St.Joseph')
('Jane', 15, 'Florida', 'St.Mary')
('John', 12, 'Tampa', 'St.Joseph', 'Jane', 15, 'Florida', 'St.Mary')
```

**Source Code:**


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2025-2029-IT-A
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stdTup.py

```
# write your code here...
# Read details of Member 1
name1, age1, address1, college1 = input().split()
member1 = (name1, int(age1), address1, college1)

# Read details of Member 2
name2, age2, address2, college2 = input().split()
member2 = (name2, int(age2), address2, college2)

# Concatenate the two tuples
combined = member1 + member2

# Print the tuples
print(member1)
print(member2)
print(combined)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| John 12 Tampa St.Joseph |
| Jane 15 Florida St.Mary |
| ('John', 12, 'Tampa', 'St.Joseph') |
| ('Jane', 15, 'Florida', 'St.Mary') |
| ('John', 12, 'Tampa', 'St.Joseph', 'Jane', 15, 'Florida', 'St.Mary') |

| Test Case - 2 |
|---|
| **User Output** |
| Ram 22 Delhi IITDelhi |
| Varun 24 Mumbai NITWarangal |
| ('Ram', 22, 'Delhi', 'IITDelhi') |
| ('Varun', 24, 'Mumbai', 'NITWarangal') |
| ('Ram', 22, 'Delhi', 'IITDelhi', 'Varun', 24, 'Mumbai', 'NITWarangal') |

| S.No: 21 | Exp. Name: ***Vowel Counter*** | Date: 2026-10-03 |
|---|---|---|

**Aim:**

Write a Python program to count the number of vowels in a given string without using control flow statements.

**Input Format:**

The program should prompt the user to enter a string.

**Output Format:**

Display the number of vowels present in the entered string.

**Source Code:**

vowel.py

```
def count_vowels(s):
        count = s.count("a") + s.count("e") + s.count("i") +
s.count("o") + s.count("u")
        count += s.count("A") + s.count("E") + s.count("I") +
s.count("O") + s.count("U")
        return count
s = input()
print(count_vowels(s))

        # Type your content here
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| BCDFGHJKL |
| 0 |

| Test Case - 2 |
|---|
| **User Output** |
| CodeTantra |
| 4 |

| Test Case - 3 |
| --- |
| **User Output** |
| Python Programming |
| 4 |

| S.No: 22 | Exp. Name: ***Key Lookup in Dictionary*** | Date: 2026-10-03 |
|---|---|---|

**Aim:**

Write a Python program that checks if a given key exists in a dictionary.

**Input Format:**

- The first line of input contains space separated strings that represent the keys of a dictionary.
- The second line contains space separated strings that represent the values of a dictionary.
- If the number of keys and values are not equal, the program should immediately print "Invalid" and terminate without taking the third input.
- Otherwise, the third line contains a string representing the key to check.

**Output Format:**

- If the number of keys and values are not equal, print "Invalid".
- Otherwise:
  - Print the dictionary in the first line.
  - Print "Exist" if the key is present in the dictionary.
  - Print "Does not Exist" if the key is not present in the dictionary.

**Note:**

- Refer to the visible test cases to strictly match the input/output format.

**Source Code:**

keyExist.py

```
keys=input().split()
values = input().split()
if len(keys)!=len(values):
        print("Invalid")
else:
        d = dict(zip(keys, values))
        key = input()
        print(d)
        if key in d:
                print("Exist")
        else:
                print("Does not Exist")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| 1 2 3 4 5 |
| A B C D E |
| 5 |
| {'1': 'A', '2': 'B', '3': 'C', '4': 'D', '5': 'E'} |
| Exist |

| Test Case - 2 |
| --- |
| **User Output** |
| 10 20 30 40 50 60 70 80 |
| Ten Twenty Thirty Forty Fifty Sixty |
| Invalid |

| Test Case - 3 |
| --- |
| **User Output** |
| John Reena Sara David |
| 24 32 18 41 |
| Jane |
| {'John': '24', 'Reena': '32', 'Sara': '18', 'David': '41'} |
| Does not Exist |

| S.No: 23 | Exp. Name: ***Add Key Value Pair to Dictionary*** | **Date: 2026-10-03** |
|---|---|---|

**Aim:**

Write a Python program that adds a new key-value pair to an existing dictionary.

**Input Format:**

- The first line contains space-separated strings that represent the keys of a dictionary.
- The second line contains space-separated strings that represent the values of the dictionary.
- The third line contains two space-separated strings for the new key and value,only if the number of keys and values match; otherwise, print "Invalid" and skip this input.

**Output Format:**

- First line should display the original dictionary before adding the new key-value pair.
- Second line should display the updated dictionary after adding the new key-value pair.

**Note:**

- If the number of keys and values does not match, display "Invalid" without taking the key-value pair as input.

**Source Code:**

keyValue.py

```
# Type Content here...
keys = input().split()
values = input().split()
if len(keys)!=len(values):
        print("Invalid")
else:
        key, value = input().split()
        d = dict(zip(keys, values))
        print(d)
        d[key] = value
        print(d)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| Mobile Laptop Adopter Headset |
| 10000 50000 2000 |
| Invalid |

| Test Case - 2 |
| --- |
| **User Output** |
| thirty fifty seventy ninety forty<br>eighty |
| 30 50 70 90 40 80 |
| four 4 |
| {'thirty': '30', 'fifty': '50', 'seventy': '70', 'ninety': '90',<br>'forty': '40', 'eighty': '80'} |
| {'thirty': '30', 'fifty': '50', 'seventy': '70', 'ninety': '90',<br>'forty': '40', 'eighty': '80', 'four': '4'} |

| S.No: 24 | Exp. Name: ***Sum of all Items in a Dictionary*** | **Date: 2026-10-03** |
|---|---|---|

**Aim:**

Write a program that allows the user to create a dictionary by entering key-value pairs. After creating the dictionary, calculate and print the sum of all values in it.

**Input Format:**

1. The first line prompts "size:" followed by an integer $size$, representing the number of key-value pairs to be added to the dictionary.
2. For each key-value pair:
   - The first line prompts "key:" followed by a string representing the $key$ on the same line.
   - The second line prompts "value:" followed by an integer representing the $value$ on the same line.

**Output Format:**

- Print "Sum:" followed by the total sum of all values without any space on the same line.

**Note:**

- Refer to the visible test cases for better understanding of input and output format.

**Source Code:**

sumOfValues.py

```
# write your code here...
n = int(input("size:"))
d = {}
for i in range(n):
        key = input("key:")
        value = int(input("value:"))
        d[key] = value
print("Sum:", sum(d.values()), sep="")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| size: |

| |
|---|
| 1 |
| key: |
| ninety |
| value: |
| 90 |
| Sum:90 |

| Test Case - 2 |
|---|
| **User Output** |
| size: |
| 4 |
| key: |
| five |
| value: |
| 5 |
| key: |
| six |
| value: |
| 6 |
| key: |
| twelve |
| value: |
| 12 |
| key: |
| sixty |
| value: |
| 60 |
| Sum:83 |

| S.No: 25 | Exp. Name: ***Sort Contents of File*** | Date: 2026-10-10 |
|---|---|---|

**Aim:**

Write a Python program that reads the contents of a text file, processes the text, sorts it, and writes the sorted data into another file.

The program should perform the following operations:

- Read the name of an input text file from the user.
- Open and read all lines from the specified file.
- Remove leading and trailing whitespace from each line.
- Convert each line to lowercase.
- Sort the lines in alphabetical order in a case-insensitive manner.
- Write the sorted lines into a new file named sorted_output.txt, with each line written on a new line.
- Display the sorted contents on the console.
- The program must handle errors appropriately.

**Input Format:**

- The input consists of a single line containing the name of the input text file.

**Output Format:**

1.If the file exists:

- Print the sorted contents of the file, one line per line.
- Create a file named sorted_output.txt containing the sorted data.

2.If the file does not exist, print:

```
File not found
```

**Note:**

- Refer to shown test cases to match with the input and output format.
- Use string methods `strip()`, `len()`, list methods `sort()`, `append()`, and file methods `open()`, `readlines()`, and `write()`.

**Source Code:**


ID: 25B01A1216

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sortLower.py

```
# write your code here...
filename = input()
try:
        file = open(filename,"r")
        lines = file.readlines()
        file.close()
        sorted_lines =[]
        for line in lines:
                line = line.strip().lower()
                if len(line)>0:
                        sorted_lines.append(line)
        sorted_lines.sort()
        output_file = open("sorted_output.txt","w")
        for line in sorted_lines:
                output_file.write(line + "\n")
        output_file.close()
        for line in sorted_lines:
                print(line)
except FileNotFoundError:
        print("File not found")
```

file1.txt

```
Banana
apple
orange
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| file1.txt |
| apple |
| banana |
| orange |

| Test Case - 2 |
| --- |

| User Output |
| --- |
| inputFile.txt |
| File not found |

| S.No: 26 | Exp. Name: ***Count the Sentences, Words and Characters in a File*** | **Date: 2026-10-10** |
|---|---|---|






**Aim:**

Write a Python program that reads a file and prints the number of lines, words, and characters present in the file.

**Input Format:**

- The input consists of a single line containing the name of the file to be read.

**Output Format:**

Print the following counts in separate lines:

- The first line will contain the number of lines in the file, prefixed with "Lines: "
- The second line will contain the number of words in the file, prefixed with "Words: "
- The third line will contain the number of characters in the file, prefixed with "Chars: "

**Source Code:**

Count.py

```
fp = input("file name:")
# write your code here
with open(fp,"r") as file:
        content = file.read()
        lines = len(content.splitlines())
        words = len(content.split())
        chars = len(content)




print("Lines:",lines)
print("Words:",words)
print("Chars:",chars)
```

input1.txt

```
Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Everything matters.
```

input2.txt

```
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
Now is better than never.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| file name: |
| input1.txt |
| Lines: 8 |
| Words: 34 |
| Chars: 228 |

| Test Case - 2 |
|---|
| **User Output** |
| file name: |
| input2.txt |
| Lines: 7 |
| Words: 51 |
| Chars: 306 |

| S.No: 27 | Exp. Name: ***Array Operations*** | Date: 2026-11-07 |
|---|---|---|

**Aim:**

Write a Python program to create an array of strings entered by the user and perform the following operations sequentially:

- Display the array.
- Append an item to the array.
- Insert an item at a specified position in the array.
- Reverse the order of items in the array.

**Input Format:**

- The first line contains space-separated string values to initialize the array.
- The second line contains a single string to append to the array.
- The third line contains an integer representing the position at which to insert the value.
- The fourth line contains a single string value to insert into the array.

**Output Format:**

- Display the created array.
- Display the array after appending the value.
- Display the array after inserting the value at the specified position.
- Display the reversed array.

**Note:** The index position to insert the value should be less than or equal to the length of the array. Otherwise, print "Invalid index".

Refer to the shown test cases to match the input and output format with the test cases.

**Source Code:**

arrayOp.py

```
# Type Content here...
arr = input().split()
arr1 = list(arr)
item = str(input())
arr.append(item)
arr2 = list(arr)
position = int(input())
insert_item = input().strip()
if 0 <= position <= len(arr):
    print(arr1)
    print(arr2)
    arr.insert(position, insert_item)
    print(arr)
    arr.reverse()
    print(arr)
else:
    print(arr1)
    print(arr2)
    print("Invalid index")
    arr.reverse()
    print(arr)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 12 23 34 45 56 67 |
| 85 |
| 2 |
| 69 |
| ['12', '23', '34', '45', '56', '67'] |
| ['12', '23', '34', '45', '56', '67', '85'] |
| ['12', '23', '69', '34', '45', '56', '67', '85'] |
| ['85', '67', '56', '45', '34', '69', '23', '12'] |

| Test Case - 2 |
|---|
| **User Output** |

| a b c d e f g h |
|---|
| k |
| 5 |
| o |
| ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h'] |
| ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'k'] |
| ['a', 'b', 'c', 'd', 'e', 'o', 'f', 'g', 'h', 'k'] |
| ['k', 'h', 'g', 'f', 'o', 'e', 'd', 'c', 'b', 'a'] |

| Test Case - 3 |
|---|
| **User Output** |
| hi hello how are you |
| fine |
| 7 |
| you |
| ['hi', 'hello', 'how', 'are', 'you'] |
| ['hi', 'hello', 'how', 'are', 'you', 'fine'] |
| Invalid index |
| ['fine', 'you', 'are', 'how', 'hello', 'hi'] |

| S.No: 28 | Exp. Name: ***Matrix Addition*** | Date: 2026-11-07 |
|---|---|---|

**Aim:**

Write a Python program to add two matrices. The program reads the dimensions and elements of two matrices and computes their sum if the dimensions are compatible. If the dimensions are incompatible, the program prints an appropriate error message.

Two matrices can be added only if they have the same number of rows and columns, i.e., $r1 = r2$ and $c1 = c2$.

**Input Format:**

- The first line contains two space-separated integers representing the number of rows ($r1$) and columns($c1$) for matrix1
- Next, $r1$ lines contain $c1$ number of integers separated by a space, representing the elements of matrix 1
- The next line contains two space-separated integers representing the number of rows ($r2$) and columns ($c2$) for matrix2
- Next, $r2$ lines contain $c2$ number of integers separated by a space, representing the elements of matrix 2

**Output Format:**

- If the matrices have compatible dimensions ($r1 == r2$ and $c1 == c2$), print the resulting matrix where each row is printed on a new line and elements are separated by spaces, otherwise print "Invalid".

**Source Code:**

matrix.py

```
# Type Content here...

r1, c1 = map(int, input().split())
matrix1 = [list(map(int, input().split())) for _ in range(r1)]
r2, c2 = map(int, input().split())
matrix2 = [list(map(int, input().split())) for _ in range(r2)]
if r1 == r2 and c1 == c2:
    for i in range(r1):
        row = []
        for j in range(c1):
            row.append(matrix1[i][j] + matrix2[i][j])
        print(*row)
else:
    print("Invalid")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| 2 3 |
| 1 2 3 |
| 4 5 6 |
| 2 3 |
| 7 8 9 |
| 0 1 2 |
| 8 10 12 |
| 4 6 8 |

| Test Case - 2 |
| --- |
| **User Output** |
| 2 2 |
| 1 2 |
| 2 3 |
| 2 3 |
| 1 2 3 |
| 0 2 3 |
| Invalid |

| S.No: 29 | Exp. Name: ***Matrix Multiplication*** | **Date: 2026-11-07** |
|---|---|---|

**Aim:**

Write a Python function **matrix_multiplication()** that performs matrix multiplication of two matrices and prints the resulting matrix. If the matrices cannot be multiplied, the function should print an appropriate error message.

**Input Format:**

- The dimensions of the first matrix ($r1$ $c1$), where $r1$ is the number of rows and $c1$ is the number of columns.
- The elements of the first matrix are provided row by row.
- The dimensions of the second matrix ($r2$ $c2$), where $r2$ is the number of rows and $c2$ is the number of columns.
- The elements of the second matrix are provided row by row.

**Output Format:**

- Print the resulting matrix after multiplication, where each element of the matrix is separated by a space; otherwise, print "Invalid".

**Note:** The number of columns of the first matrix must be equal to the number of rows of the second matrix for matrix multiplication.

**Source Code:**

matrixmul.py

```
# define the function
def matrix_multiplication():
        r1, c1 = map(int, input().split())
        matrix1 = [list(map(int, input().split())) for _ in range(r1)]

        r2, c2 = map(int, input().split())
        matrix2 = [list(map(int, input().split())) for _ in range(r2)]

        if c1 != r2:
                print("Invalid")
                return

        result = [[0] * c2 for _ in range(r1)]

        for i in range(r1):
                for j in range(c2):
                        for k in range(c1):
                                result[i][j] += matrix1[i][k] *
matrix2[k][j]

        for row in result:
                print(*row)

# Call the function to run matrix multiplication




# Call the function to run matrix multiplication
matrix_multiplication()
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 2 3 |
| 1 2 3 |
| 4 5 6 |
| 3 4 |

| |
|---|
| 1 2 3 4 |
| 5 6 7 8 |
| 9 0 1 2 |
| 38 14 20 26 |
| 83 38 53 68 |

| Test Case - 2 |
|---|
| **User Output** |
| 2 2 |
| -3 4 |
| 4 -7 |
| 2 2 |
| -4 5 |
| -6 7 |
| -12 13 |
| 26 -29 |

| Test Case - 3 |
|---|
| **User Output** |
| 2 2 |
| 6 12 |
| -3 7 |
| 3 2 |
| 12 23 34 |
| 12 34 54 |
| 4 26 19 |
| Invalid |

| S.No: 30 | Exp. Name: ***Transpose of Matrix*** | Date: 2026-11-06 |
|---|---|---|

**Aim:**

Write a Python program to find the transpose of a matrix.

**Input Format:**

- The first line contains two space-separated integers representing the number of rows ($R$) and columns ($C$) for the matrix.
- Next, $R$ lines contain $C$ number of integers separated by a space, representing the elements of the matrix.

**Output Format:**

- Print the transpose of the matrix, with each row on a new line and elements space-separated.

**Source Code:**

transpose.py

```
r, c =map(int, input().split())
a =[]
for i in range(r):
        a.append(list(map(int, input().split())))
for j in range(c):
        row =[]
        for i in range(r):
                row.append(a[i][j])
        print(*row)
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 2 4 |
| 10 20 30 40 |
| 12 23 45 52 |
| 10 12 |
| 20 23 |
| 30 45 |
| 40 52 |

| Test Case - 2 |
| --- |
| **User Output** |
| 3 5 |
| -10 -20 -30 -40 -50 |
| -11 -22 -33 -44 -55 |
| -13 -15 -17 -18 -19 |
| -10 -11 -13 |
| -20 -22 -15 |
| -30 -33 -17 |
| -40 -44 -18 |
| -50 -55 -19 |

| S.No: 31 | Exp. Name: ***Geometry Class*** | Date: 2026-11-06 |
|---|---|---|






**Aim:**

Write a Python program that defines a base class **Shape** with two methods, **area** and **perimeter**, that are meant to be overridden by subclasses. Define three subclasses: **Circle**, **Triangle**, and **Square**, each of which implements the **area** and **perimeter** methods. The program should then create instances of these subclasses based on user input and print their areas and perimeters, formatted to two decimal places.

**Input Format:**

- The first line contains a single float representing the radius of a circle.
- The second line contains three space-separated floats representing the sides of a triangle.
- The third line contains a single float representing the side length of a square.

**Output Format:**

- Print the area and perimeter of the circle on one line, separated by a space, formatted to two decimal places.
- Print the area and perimeter of the triangle on the next line, separated by a space, formatted to two decimal places.
- Print the area and perimeter of the square on the last line, separated by a space, formatted to two decimal places.

**Note:**

- Triangle sides must satisfy the triangle inequality theorem: the sum of any two sides must be greater than the third side, ensuring the formation of a valid triangle.
- Area of traingle $ABC = \sqrt{[s \times (s-a) \times (s-b) \times (s-c)]}$
- $s = \frac{(side1 + side2 + side3)}{2}$

**Source Code:**

geometry1.py

```
import math

# Shape class
class Shape:
        def area(self):
                pass
        def perimeter(self):
                pass
# Circle subclass
class Circle(Shape):
        def __init__(self, radius):
                self.radius = radius
        def area(self):
                return math.pi * self.radius*self.radius
        def perimeter(self):
                return 2*math.pi*self.radius
# Triangle subclass
class Triangle(Shape):
        def __init__(self,a,b,c):
                self.a=a
                self.b=b
                self.c=c
        def area(self):
                s =(self.a + self.b+self.c)/2
                return math.sqrt(s*(s-self.a)*(s-self.b)*(s-self.c))
        def perimeter(self):
                return self.a+self.b+self.c
# Square subclass
class Square(Shape):
        def __init__(self, side):
                self.side =side
        def area(self):
                return self.side*self.side
        def perimeter(self):
                return 4* self.side
r = float(input())
a,b,c =map(float,  input().split())
s = float(input())
circle = Circle(r)
triangle =Triangle(a,b,c)
square =Square(s)
print(f"{circle.area():.2f} {circle.perimeter():.2f}")
print(f"{triangle.area():.2f} {triangle.perimeter():.2f}")
print(f"{square.area():.2f} {square.perimeter():.2f}")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| 12 |
| 4 5 6 |
| 4 |
| 452.39 75.40 |
| 9.92 15.00 |
| 16.00 16.00 |

| Test Case - 2 |
| --- |
| **User Output** |
| 5.0 |
| 3.0 4.0 5.0 |
| 7.5 |
| 78.54 31.42 |
| 6.00 12.00 |
| 56.25 30.00 |

| S.No: 32 | Exp. Name: ***Slicing, Boolean Indexing of Array*** | **Date: 2026-11-07** |
|---|---|---|

**Aim:**

Write a python program to demonstrate basic slicing, integer and Boolean indexing of array using **NumPy** operations.

**Input & Output Format:**

- First line: Two integers specifying number of rows (rows) and columns (cols) of the array (space separated).
- Next rows lines: Each line contains cols integers (space separated) representing the elements of the array row-wise.
- Then, display the NumPy array after constructing it from the provided input elements.
- Next line: Two integers (start and end) specifying the range of rows to slice from the array (space separated).
- Then, display the sliced portion of the array based on the start and end indices. If the indices are invalid ((i.e., start $<$ 0 or start $>=$ rows or end $<$ 0 or end $>$ rows), output "Invalid".
- Next line: An integer (index) specifying the row position to access an element using integer indexing.
- Display the row in the array at the specified index. If the index is invalid, output "Invalid"((i.e., not in the range 0 to rows - 1).
- Next line: A string of space-separated integers (0 or 1) representing boolean indices for selecting rows from the array. If the number of boolean values is not equal to the number of rows, output "Invalid".
- Display the boolean index array used for selection followed by elements from the array that correspond to True values in the boolean index array.

**Source Code:**

array_operations.py

```
# write your code here...
import numpy as np

rows, cols = map(int, input().split())
elements = []
for _ in range(rows):
    elements.extend(list(map(int, input().split())))
arr = np.array(elements).reshape(rows, cols)
print(arr)

start, end = map(int, input().split())
if start < 0 or start >= rows or end <= 0 or end > rows or start > end:
    print("Invalid")
else:
    print(arr[start:end])

idx = int(input())
if 0 <= idx < rows:
    print(arr[idx])
else:
    print("Invalid")

bool_inputs = list(map(int, input().split()))
if len(bool_inputs) != rows:
    print("Invalid")
else:
    bool_arr = np.array(bool_inputs, dtype=bool)
    print(bool_arr)
    print(arr[bool_arr])
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| 3 4 |
| 1 2 3 4 |
| 5 6 7 8 |
| 9 10 11 12 |
| [[ 1  2  3  4] |

```
 [ 5  6  7  8]
 [ 9 10 11 12]]
0 2
[[1 2 3 4]
 [5 6 7 8]]
1
[5 6 7 8]
1 0 1
[ True False  True]
[[ 1  2  3  4]
 [ 9 10 11 12]]
```

| Test Case - 2 |
|---|
| **User Output** |
| 2 3 |
| 1 2 3 |
| 4 5 6 |
| [[1 2 3] |
| [4 5 6]] |
| 1 4 |
| Invalid |
| 3 |
| Invalid |
| 1 0 |
| [ True False] |
| [[1 2 3]] |

| Test Case - 3 |
|---|
| **User Output** |
| 3 2 |
| 1 2 |
| 3 4 |
| 5 6 |
| [[1 2] |
| [3 4] |
| [5 6]] |
| -1 2 |
| Invalid |

| |
|---|
| -1 |
| Invalid |
| 0 0 0 |
| [False False False] |
| [] |

| Test Case - 4 |
|---|
| **User Output** |
| 0 0 |
| [] |
| 0 0 |
| Invalid |
| 0 |
| Invalid |
| 0 |
| Invalid |

| S.No: 33 | Exp. Name: ***Scatter Plot using Pandas Dataframe*** | **Date: 2026-11-07** |
|---|---|---|

**Aim:**

Create a scatter plot to visualize the relationship between two columns from a dataset represented as a dictionary. Convert the dictionary into a Pandas DataFrame for this purpose.

**Input Format:**

- Prompt the user to input two column names (strings) separated by space.

**Output Format:**

- Display a scatter plot showing the relationship between the selected columns.
- Ensure the plot includes appropriate labels, title, legend, grid, and customization (such as color and marker style).

**Plot Settings:**

- Use blue color to set the color of the markers
- Marker Style: Use marker='o' to set the marker style
- Use "Scatter" as label for the scatter plot in the legend.
- Set "Scatter Plot" as the title of the plot.
- Use column1 and column2 as the labels for the x-axis and y-axis, respectively.
- Display the legend.
- Display the grid on the plot for better visualization.

**Note:** Refer to the data set that has been provided to you in the editor.

**Source Code:**

scatter.py

```
import pandas as pd
import matplotlib.pyplot as plt

# Given DataFrame
std_marks = {
    'A': [10, 20, 30, 40, 50, 60, 70, 80, 90, 100],
    'B': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'C': [11, 22, 33, 44, 55, 66, 77, 88, 99, 101],
    'D': [103, 105, 109, 114, 125, 135, 149, 154, 162, 185],
    'E': [52, 86, 75, 41, 32, 71, 68, 94, 73, 69]
}
# Construct a Pandas DataFrame using the provided item data
df = pd.DataFrame(std_marks)

# Prompt the user to input two column names (strings) seperated by space
col1, col2 = input().split()

# Filter the DataFrame to include only relevant columns for
visualization
if col1 in df.columns and col2 in df.columns:
    plt.scatter(df[col1], df[col2], color='blue', marker='o',
label='Scatter')
    plt.xlabel(col1)
    plt.ylabel(col2)
    plt.title('Scatter plot')
    plt.legend()
    plt.grid(True)
    plt.show()
else:
    print("Invalid")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| A C |
| data:img/png;base64,iVBORw0KGgoAAAANSUhEUgAAAoAAAAHgCAYAAAA10dzkAAAAOX |

| Test Case - 2 |
| --- |
| **User Output** |
| B D |
| data:img/png;base64,iVBORw0KGgoAAAANSUhEUgAAAoAAAAHgCAYAAAA10dzkAAAAO |

| S.No: 34 | Exp. Name: ***Numpy Array*** | Date: 2026-11-06 |
|---|---|---|






**Aim:**

Write a Python program that prompts the user to enter integer elements for a **NumPy** array. The program should create the **NumPy** array using the entered elements and display it.

**Input Format:**

- A single line of space-separated integer values entered by the user.

**Output Format:**

- The **NumPy** array created from the user's input.

**Note:**

- Refer to the sample test cases for better understanding regarding input and output format.

**Source Code:**

arrayNumpy.py

```
import numpy as np
a = list(map(int, input().split()))
arr = np.array(a)
print(arr)
#type your code here...
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 1 2 3 4 5 |
| [1 2 3 4 5] |

| Test Case - 2 |
|---|
| **User Output** |
| 10 20 30 40 50 60 70 80 90 |
| [10 20 30 40 50 60 70 80 90] |

| S.No: 35 | Exp. Name: ***Numpy Array Operations*** | Date: 2026-11-06 |
|---|---|---|

**Aim:**

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

**Input Format:**

- The first line contains two space-separated integers, $r$ and $c$, representing the number of rows and the number of columns.
- The next $r$ lines contain $c$ space-separated integers representing the elements of the array, entered row by row.

**Output Format:**

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

**Note:** Use `reshape()` function to reshape the input array with the specified number of rows and columns.

**Source Code:**

numpyarr.py

```
import numpy as np
r, c =map(int, input().split())
a =[]
for i in range(r):
        a.extend(map(int, input().split()))
arr =np.array(a).reshape(r,c)
print(arr)
print(arr.ndim)
print(arr.shape)
print(arr.size)
# write your code here...
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |
| 3 4 |
| 1 2 3 4 |
| 5 6 7 8 |
| 9 10 11 12 |
| [[ 1  2  3  4] |
| [ 5  6  7  8] |
| [ 9 10 11 12]] |
| 2 |
| (3, 4) |
| 12 |

| Test Case - 2 |
| --- |
| **User Output** |
| 0 0 |
| [] |
| 2 |
| (0, 0) |
| 0 |

| S.No: 36 | Exp. Name: ***Numpy Array Statistics*** | Date: 2026-11-06 |
|---|---|---|









**Aim:**

Write a Python program that prompts the user to enter elements for a NumPy array. The program should compute and print the minimum value, maximum value and sum of elements of the array using NumPy functions.

**Input Format:**

- The program expects the user to enter elements of the array separated by spaces.

**Output Format:**

- First, print the original NumPy array as integers.
- Next, print the minimum value found in the array as an integer.
- Then, display the maximum value present in the array as an integer.
- Finally, show the sum of all elements within the array as an integer.

**Source Code:**

arrayStat.py

```
import numpy as np
a = list(map(int, input().split()))
arr = np.array(a)
print(arr)
print(np.min(arr))
print(np.max(arr))
print(np.sum(arr))

# write your code here...
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| **User Output** |
| 1 2 3 4 5 6 7 8 9 |
| [1 2 3 4 5 6 7 8 9] |
| 1 |
| 9 |
| 45 |

| Test Case - 2 |
| --- |
| **User Output** |
| -10 -20 -30 -40 -50 |
| [-10 -20 -30 -40 -50] |
| -50 |
| -10 |
| -150 |

| S.No: 37 | Exp. Name: ***Numpy, pandas operations using dictionary*** | **Date: 2026-11-07** |
|---|---|---|

**Aim:**

Convert the given dataset into a Pandas DataFrame representing students (A, B, C, D, E) and their marks in ten different subjects. Each student's data is structured as follows:

- Keys represent student names (A, B, C, D, E).
- Values are lists containing the marks for ten subjects respectively.

Define a function that allows the user to perform a custom data selection operation on this DataFrame. The function should:

- Take user input for the student's name (column) and a minimum marks threshold (condition_value).
- Apply the head() function to display the initial rows of the DataFrame.
- Filter and display the rows (students) whose marks in the specified subject (column) are greater than the provided threshold (condition_value).

**Input Format:**

- Prompt the user to input the student's name (A, B, C, D, E).
- Prompt the user to input the minimum marks they want to filter by in the selected subject.

**Output Format:**

- Display the output of the head() function applied to the Pandas DataFrame.
- Print the marks of the specified student that meet the specified criteria.

**Source Code:**


ID: 25B01A1216

Shri Vishnu Engineering College for Women (Autonomous)

stdMarksDf.py

```
import pandas as pd

# Predefined dictionary with at least five keys, each representing a
student's subject marks
std_marks = {
    'A': [10, 20, 30, 40, 50, 60, 70, 80, 90, 100],
    'B': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'C': [11, 22, 33, 44, 55, 66, 77, 88, 99, 101],
    'D': [103, 105, 109, 114, 125, 135, 149, 154, 162, 185],
    'E': [52, 86, 75, 41, 32, 71, 68, 94, 73, 69]
}
# Convert dictionary to Pandas DataFrame
df = pd.DataFrame(std_marks)

# Prompt user for student name
student_name = input().strip()

# Prompt user for minimum marks
min_marks = float(input())

# Function to display marks greater than the provided threshold.
def custom_selection(df, col, threshold):
    print(df.head())
    if col in df.columns:
        return df[col][df[col] > threshold]
    else:
        return pd.Series(dtype=int)

# Call the function and assign to selected_data
selected_data = custom_selection(df, student_name, min_marks)

# Display selected data
if not selected_data.empty:
    print(selected_data)
else:
    print("Invalid")
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
| --- |
| **User Output** |

```
A
35
    A  B   C    D   E
0  10  1  11  103  52
1  20  2  22  105  86
2  30  3  33  109  75
3  40  4  44  114  41
4  50  5  55  125  32
3    40
4    50
5    60
6    70
7    80
8    90
9   100
Name: A, dtype: int64
```

**Test Case - 2**

**User Output**

```
E
100
    A  B   C    D   E
0  10  1  11  103  52
1  20  2  22  105  86
2  30  3  33  109  75
3  40  4  44  114  41
4  50  5  55  125  32
Invalid
```

**Test Case - 3**

**User Output**

```
C
62.83
    A  B   C    D   E
0  10  1  11  103  52
1  20  2  22  105  86
2  30  3  33  109  75
3  40  4  44  114  41
4  50  5  55  125  32
```

```
5     66
6     77
7     88
8     99
9    101
Name: C, dtype: int64
```